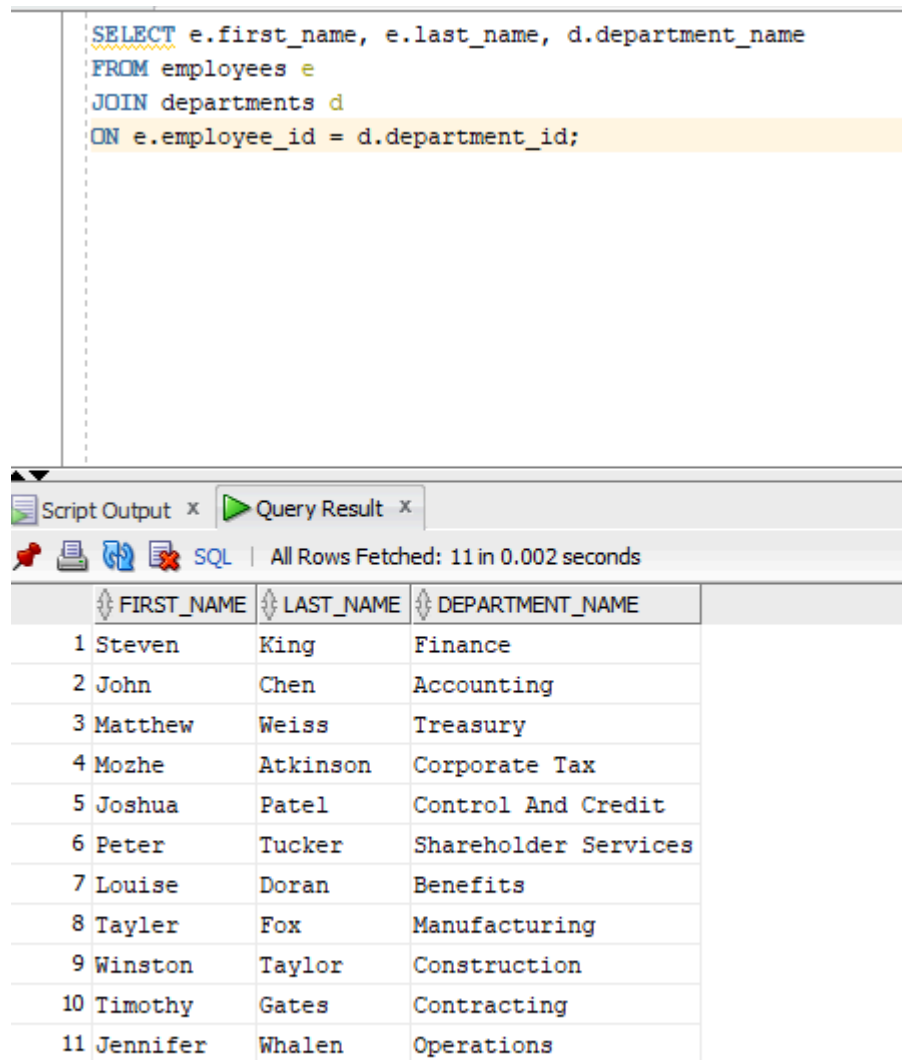


Oracle SQL Code Review & Query Logic Validation

Query Review 1

```
SELECT e.first_name, e.last_name, d.department_name
FROM employees e
JOIN departments d
ON e.employee_id = d.department_id;
```



The screenshot shows an Oracle SQL Developer interface. The top pane displays the SQL query: `SELECT e.first_name, e.last_name, d.department_name FROM employees e JOIN departments d ON e.employee_id = d.department_id;`. The bottom pane shows the query results in a table with 11 rows. The table has columns: FIRST_NAME, LAST_NAME, and DEPARTMENT_NAME. The results are as follows:

	FIRST_NAME	LAST_NAME	DEPARTMENT_NAME
1	Steven	King	Finance
2	John	Chen	Accounting
3	Matthew	Weiss	Treasury
4	Mozhe	Atkinson	Corporate Tax
5	Joshua	Patel	Control And Credit
6	Peter	Tucker	Shareholder Services
7	Louise	Doran	Benefits
8	Tayler	Fox	Manufacturing
9	Winston	Taylor	Construction
10	Timothy	Gates	Contracting
11	Jennifer	Whalen	Operations

Syntax Validation

The SQL syntax is valid.

The query executes successfully and returns rows.

Result Analysis

The query returned employee names with department names. However, the returned data is not reliable because the join condition is incorrect.

Business Logic Issue

The query joins:

`e.employee_id = d.department_id`

This is not a valid business relationship.

`employee_id` identifies an employee, while `department_id` identifies a department.

These two columns should not be matched together. Any matching rows are accidental, not meaningful.

Correct Query

```
SELECT e.first_name,  
       e.last_name,  
       d.department_name  
FROM employees e  
JOIN departments d  
ON e.department_id = d.department_id;
```

```

SELECT e.first_name,
       e.last_name,
       d.department_name
FROM employees e
JOIN departments d
ON e.department_id = d.department_id;

```

Script Output x Query Result x			
SQL Fetched 50 rows in 0.007 seconds			
	FIRST_NAME	LAST_NAME	DEPARTMENT_NAME
1	Jennifer	Whalen	Administration
2	Pat	Fay	Marketing
3	Michael	Hartstein	Marketing
4	Sigal	Tobias	Purchasing
5	Karen	Colmenares	Purchasing
6	Shelli	Baida	Purchasing
7	Den	Raphaely	Purchasing
8	Alexander	Khoo	Purchasing
9	Guy	Himuro	Purchasing
10	Susan	Mavris	Human Resources
11	Kevin	Feeney	Shipping
12	Jean	Fleaur	Shipping
13	Adam	Fripp	Shipping
14	Timothy	Gates	Shipping
15	Ki	Gee	Shipping
16	Girard	Geoni	Shipping
17	Douglas	Grant	Shipping
18	Vance	Jones	Shipping
19	Payam	Kaufling	Shipping
20	Renske	Ladwig	Shipping
21	Britney	Everett	Shipping
22	Jennifer	Billy	Shipping

Reasoning

The correct relationship is between the employee's department ID and the department table's department ID.

`employees.department_id = departments.department_id`

This uses the foreign key relationship correctly and returns the real department for each employee.

Query Review 2

```
SELECT e.first_name, e.last_name, j.job_title
```

```
FROM employees e
```

```
JOIN jobs j
```

```
ON e.employee_id = j.job_id;
```

The screenshot shows the Oracle SQL Developer interface. The top pane, titled 'Query Builder', contains the following SQL query:

```
SELECT e.first_name, e.last_name, j.job_title
FROM employees e
JOIN jobs j
ON e.employee_id = j.job_id;
```

The bottom pane, titled 'Script Output', shows the execution of the query and the resulting error message:

```
ORA-01722: invalid number

https://docs.oracle.com/error-help/db/ora-01722/01722.00000 - "unable to convert string value containing %s to a number: %s"
*Document: YES
*Cause: The attempted conversion of a character string for column or
expression to a number failed because the character string is not
a valid numeric literal. Only numeric fields or character fields
containing numeric data can be used in arithmetic functions or
expressions. Only numeric fields can be added to or subtracted from
dates. If "UNISTR" appears in the error message, the value is
not compatible with the national character set and cannot be
represented directly.
*Action: Use the LIKE expression to identify the problematic value. Ensure
that it contains only digits, a sign, a decimal separator, and
the character "E" or "e", and retry the operation.
*Params: 1) invalid_char
the character string that is being converted to a number or
UNISTR (character string) if the character is not compatible
with the database character set.
2) column_or_expression: The column or expression from where the
invalid character comes.
```

Syntax Validation

The SQL syntax is written correctly, but the query fails during execution.

Error Returned

ORA-01722: invalid number

Result Analysis

The query does not return valid results because Oracle attempts to compare a numeric column with a character column.

Business Logic Issue

The join condition is incorrect:

`e.employee_id = j.job_id`

`employee_id` is a number, for example:

100, 101, 102

But `job_id` is text, for example:

AD_PRES, IT_PROG, SA_REP

This causes a data type mismatch and also does not represent the correct relationship between employees and jobs.

Correct Query

```
SELECT e.first_name,  
       e.last_name,  
       j.job_title  
FROM employees e  
JOIN jobs j  
ON e.job_id = j.job_id;
```

Worksheet		Query Builder																																																																																													
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Reasoning

The correct business relationship is:

employees.job_id = jobs.job_id

Each employee has a **job_id**, and the **jobs** table stores the details of each job. This query correctly shows each employee with their actual job title.

Query Review 3

```
SELECT e.first_name, d.department_name
FROM employees e
LEFT JOIN departments d
ON e.department_id = d.department_id
WHERE d.department_name = 'Sales';
```

Worksheet

Query Builder

```
SELECT e.first_name, d.department_name
FROM employees e
LEFT JOIN departments d
ON e.department_id = d.department_id
WHERE d.department_name = 'Sales';
```

Script Output x

Query Result x

    SQL | All Rows Fetched: 34 in 0.005 seconds

	FIRST_NAME	DEPARTMENT_NAME
1	John	Sales
2	Karen	Sales
3	Alberto	Sales
4	Gerald	Sales
5	Eleni	Sales
6	Peter	Sales
7	David	Sales
8	Peter	Sales
9	Christopher	Sales
10	Nanette	Sales
11	Oliver	Sales
12	Janette	Sales
13	Patrick	Sales
14	Allan	Sales
15	Lindsey	Sales
16	Louise	Sales
17	Sarath	Sales
18	Clara	Sales
19	Danielle	Sales
20	Mattea	Sales
21	David	Sales
22	Sundar	Sales

Syntax Validation

The SQL syntax is valid.

The query executes successfully and returns employees from the Sales department.

Result Analysis

The query returned Sales department employees. The output is correct for showing only Sales employees.

However, the query design is not ideal because it uses a **LEFT JOIN** but then filters the right table in the **WHERE** clause.

Business Logic Issue

This condition:

```
WHERE d.department_name = 'Sales'
```

removes all rows where **d.department_name** is **NULL**.

That means the **LEFT JOIN** no longer behaves like a real left join. It behaves like an **INNER JOIN**.

Better Correct Query

```
SELECT e.first_name,  
       d.department_name  
FROM employees e  
JOIN departments d  
ON e.department_id = d.department_id  
WHERE d.department_name = 'Sales';
```


Worksheet

Query Builder

```

SELECT e.first_name,
       d.department_name
FROM employees e
JOIN departments d
ON e.department_id = d.department_id
WHERE d.department_name = 'Sales';

```

Script Output x

Query Result x

SQL | All Rows Fetched: 34 in 0.006 seconds

	FIRST_NAME	DEPARTMENT_NAME
1	John	Sales
2	Karen	Sales
3	Alberto	Sales
4	Gerald	Sales
5	Eleni	Sales
6	Peter	Sales
7	David	Sales
8	Peter	Sales
9	Christopher	Sales
10	Nanette	Sales
11	Oliver	Sales
12	Janette	Sales
13	Patrick	Sales
14	Allan	Sales
15	Lindsey	Sales
16	Louise	Sales
17	Sarath	Sales
18	Clara	Sales
19	Danielle	Sales
20	Mattea	Sales
21	David	Sales

Reasoning

If the business requirement is:

Show only employees who work in the Sales department.

Then **INNER JOIN** is better because only matching employees and departments are needed.

The query is clearer, easier to maintain, and avoids misleading future developers.

Query Review 4

```
SELECT e.first_name, m.first_name AS manager_name  
FROM employees e  
JOIN employees m  
ON e.employee_id = m.manager_id;
```

Worksheet

Query Builder

```

SELECT e.first_name, m.first_name AS manager_name
FROM employees e
JOIN employees m
ON e.employee_id = m.manager_id;

```

Script Output x

Query Result x

SQL | Fetched 50 rows in 0.004 seconds

	FIRST_NAME	MANAGER_NAME
1	Eleni	Ellen
2	Alberto	Sundar
3	Adam	Mozhe
4	Alexander	David
5	Neena	Hermann
6	Den	Shelli
7	Alberto	Amit
8	Gerald	Elizabeth
9	Shanta	Sarah
10	John	David
11	Adam	Laura
12	Gerald	Harrison
13	Adam	Alexis
14	Adam	Anthony
15	Steven	Gerald
16	John	Nanette
17	Nancy	John
18	Payam	Kelly
19	Den	Karen
20	Kevin	Curtis
21	Steven	Lex
22	Adam	Tulia

Syntax Validation

The SQL syntax is valid.

The query executes successfully and returns rows.

Result Analysis

The query returned **50 rows**. However, the result is not showing the correct employee-manager relationship.

This result is misleading because the query is matching an employee's ID with another employee's `manager_id`.

Business Logic Issue

The join condition is incorrect:

ON e.employee_id = m.manager_id

This means:

Show employees who are managers, and show the employees who report to them.

But selected columns make it look like:

employee name | manager name

Actually, in query:

- `e` represents the manager.
- `m` represents the employee working under that manager.

So the aliases are reversed and confusing.

Correct Query

If the requirement is:

Show each employee with their manager name.

```
SELECT e.first_name AS employee_name,  
       m.first_name AS manager_name  
FROM employees e  
JOIN employees m  
ON e.manager_id = m.employee_id;
```

Worksheet		Query Builder																																																																					
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Why This Is Correct

In the HR schema, the employee table has a self-relationship:

employees.manager_id = employees.employee_id

This means:

- `e.manager_id` stores the employee's manager ID.
- `m.employee_id` stores the manager's employee ID.

So the correct relationship is:

ON `e.manager_id = m.employee_id`

Not:

ON `e.employee_id = m.manager_id`

Reason

Each employee stores their manager ID in `e.manager_id`.